Reproduction of Nakagawa, Yasuhara & Baba (ISPSD 1990)

"New 500 V Output Device Structures for Thin Silicon Layer on Silicon Dioxide Film"
Proc. ISPSD 1990, Tokyo, pp. 97–101

Independent derivation, simulation and figure-by-figure comparison

Abstract

A thin (20 µm) silicon layer on a buried silicon-dioxide film (BOX) over a grounded substrate forms a lateral high-voltage diode. When the silicon fully depletes, more than half of the applied 500 V is forced across the buried oxide, and a shallow N donor layer (dose S_D) shields the bulk silicon so avalanche is delayed. We derive the electrostatics, solve the Poisson PDE in 1-D (drain symmetry axis) and 2-D (cross-section), add a Chynoweth avalanche integral, and compare figure-by-figure with the paper. Figures 4, 5 and 6 (structure B, the optimized device) reproduce quantitatively; Fig. 7 (structure A) is a 2-D effect the vertical model captures only in trend.

Key results (model vs. paper)

Quantity	Model	Paper
Oxide voltage share ($t_{ox} = 3\,\mu m$)	54–60 %	> 50 % (Fig. 4)
Peak BV, $t_{ox} = 1\,\mu m$	401 V	$\sim$380 V (Fig. 6)
Peak BV, $t_{ox} = 2\,\mu m$	479 V	$\sim$480 V (Fig. 6)
Peak BV, $t_{ox} = 3\,\mu m$	565 V	$\sim$580 V (Fig. 6)
RESURF optimum dose	$\sim 10^{12}$	$\sim 10^{12}$ cm^{-2}
Structure A needs thicker oxide	yes (2-D)	5 µm vs 3 µm (Fig. 7)

1 Theory

1.1 The device and the key idea

Under high reverse bias the thin silicon layer *fully depletes*, and a large fraction of the applied voltage is forced across the buried oxide — so the device blocks ≥ 500 V with only 20 µm of silicon. The ionized (positive) donors of the shallow N layer at the Si/BOX interface "shield" the bulk silicon: they let the field be large in the oxide while keeping it low in the silicon, delaying avalanche.

1.2 One-dimensional vertical model (drain symmetry axis AA', structure B)

Take y downward, $y = 0$ at the top of the silicon; the drain (top) is at the applied voltage V and the substrate (below the oxide) is grounded. Under full depletion the charge density is $\rho = qN$ (donors positive) and Poisson's equation is

$$\frac{d}{dy}\left(\varepsilon \frac{d\phi}{dy}\right) = -qN(y), \qquad E = -\frac{d\phi}{dy}. \tag{1}$$

The normal displacement $D = \varepsilon E$ is continuous across the Si/oxide interface, so the field jumps *up* into the oxide:

$$E_{ox} = \frac{\varepsilon_{si}}{\varepsilon_{ox}} E_{si} \approx \frac{11.7}{3.9} E_{si} \approx 3\, E_{si}. \tag{2}$$

Modelling the shallow layer as an areal sheet qS_D, crossing it upward drops the displacement by qS_D, so the field in the *bulk* silicon is

$$\boxed{E_{si,bulk} = \frac{D_{ox} - qS_D}{\varepsilon_{si}}} \tag{3}$$

which *decreases* as S_D rises: the donor sheet shields the silicon (exactly the Fig. 5 ordering $D > C > B > A$). Integrating E over each region, the oxide share V_{ox}/V grows with S_D and t_{ox}; at $t_{\mathrm{ox}} = 3\,\mu\mathrm{m}$, $S_D = 1.5 \times 10^{12}\,\mathrm{cm}^{-2}$, $V = 500\,\mathrm{V}$ the model gives $V_{\mathrm{ox}}/V \approx 0.60$ — reproducing Fig. 4's "more than half across the oxide."

1.3 Avalanche breakdown and the RESURF optimum

Avalanche occurs in the silicon (the oxide does not ionize). With the Chynoweth ionization coefficient for silicon,

$$\alpha(E) = A\, e^{-B/E}, \qquad A = 7.03 \times 10^{5}\,\mathrm{cm}^{-1}, \quad B = 1.231 \times 10^{6}\,\mathrm{V\,cm}^{-1}, \tag{4}$$

breakdown occurs when the ionization integral along AA' reaches unity:

$$I(V) = \int_{\mathrm{Si}} \alpha\big(|E(y)|\big)\, dy = 1 \implies V = BV. \tag{5}$$

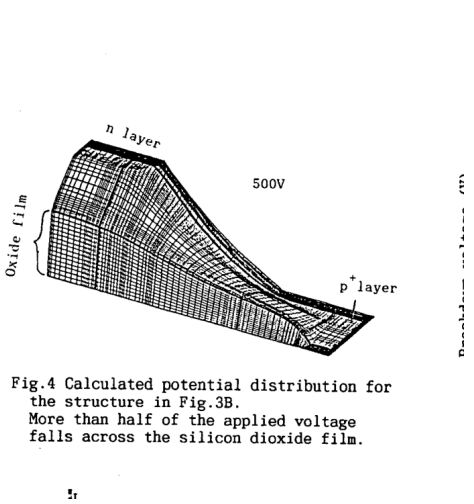
Two competing effects set $BV(S_D)$: too little dose gives poor shielding and an early avalanche; too much dose over-shields and the corner/oxide field grows. So $BV(S_D)$ rises, peaks near the RESURF charge ($\sim 1 \times 10^{12}\,\mathrm{cm}^{-2}$), then falls — the shape of Fig. 6. A thicker oxide shares more voltage and raises the peak BV.

1.4 Why structure A needs a thicker oxide (a 2-D effect)

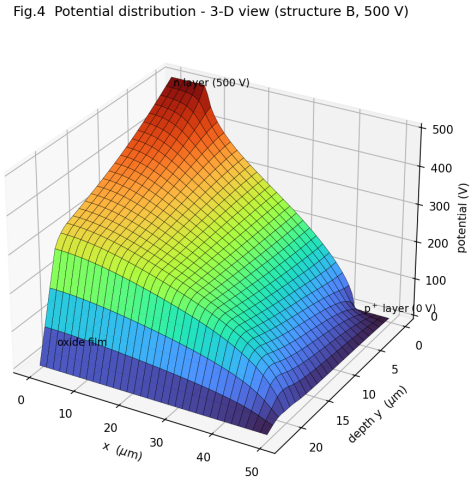
Structures A and B share the *same* vertical stack, so a purely vertical (1-D) or mirror-symmetric 2-D model cannot separate them. In structure B the drain sits directly over the oxide at the symmetry axis (most complete vertical field-sharing, breakdown point #1 under the drain); in structure A the drain is displaced and its critical point couples less completely to the grounded substrate, so A needs a 5 µm oxide where B needs 3 µm. The vertical model represents the favourable B-type sharing and therefore over-predicts A.

Assumptions. $L_{\mathrm{si}} = 20\,\mu\mathrm{m}$; $t_{\mathrm{ox}} = 1$–$5\,\mu\mathrm{m}$ (swept); shallow-layer thickness $w_n = 2\,\mu\mathrm{m}$ ("a few microns"); background n^- doping $N_{D0} = 1 \times 10^{14}\,\mathrm{cm}^{-3}$ (not stated in the paper; chosen to centre the Fig. 6 RESURF peak; standard for a 500 V-class 20 µm drift layer); $\varepsilon_{\mathrm{si}} = 11.7$, $\varepsilon_{\mathrm{ox}} = 3.9$.

2 Figure-by-figure comparison



(a) Paper (original)



(b) This reproduction (3-D)

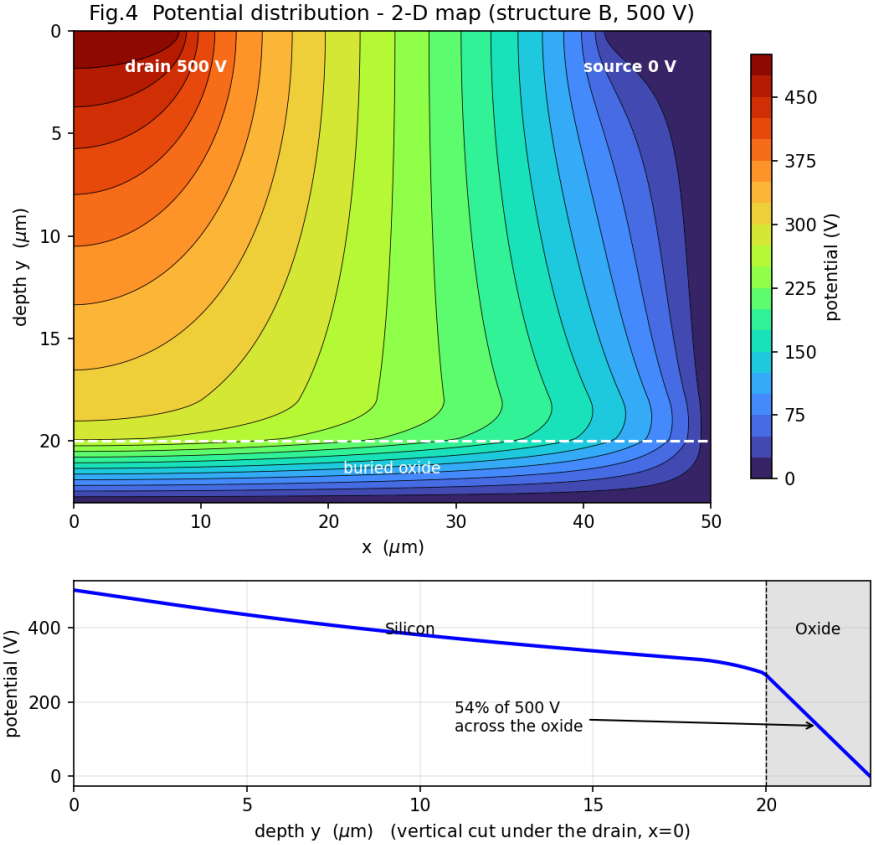


Figure 1: **Fig. 4 — potential distribution (structure B, 500 V).** Both show a 500 V plateau at the n-layer/drain sloping to the p^+ side, with a steep face over the oxide film — the >half-voltage drop across the buried oxide. The 2-D map (bottom) quantifies it: 54 % of 500 V falls across the oxide under the drain.

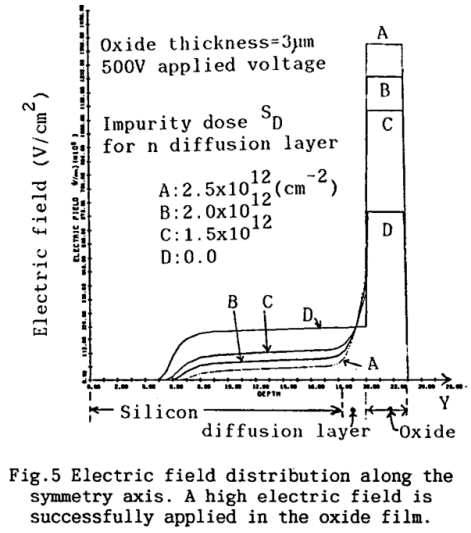
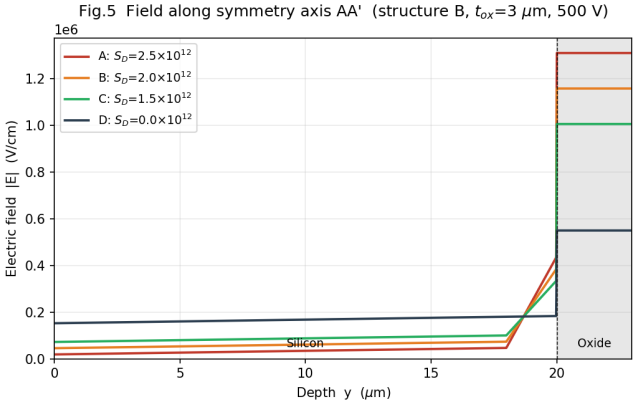


Fig.5 Electric field distribution along the symmetry axis. A high electric field is successfully applied in the oxide film.

(a) Paper (original)



(b) This reproduction

Figure 2: **Fig. 5 — electric field along the symmetry axis** AA' ($t_{\mathrm{ox}} = 3\,\mu\mathrm{m}$, $500\,\mathrm{V}$). In the silicon the field is low and ordered $D > C > B > A$ (more dose ⇒ more shielding); it jumps $\sim 3\times$ at the Si/oxide interface and is constant across the charge-free oxide. The paper's oxide bars (A highest, D lowest) match the model's oxide field.

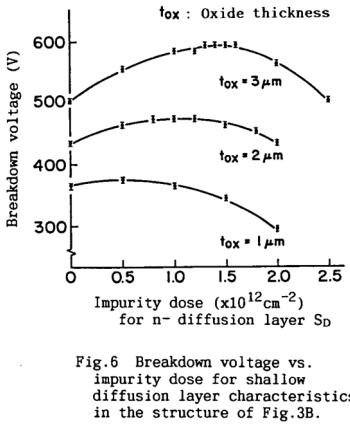
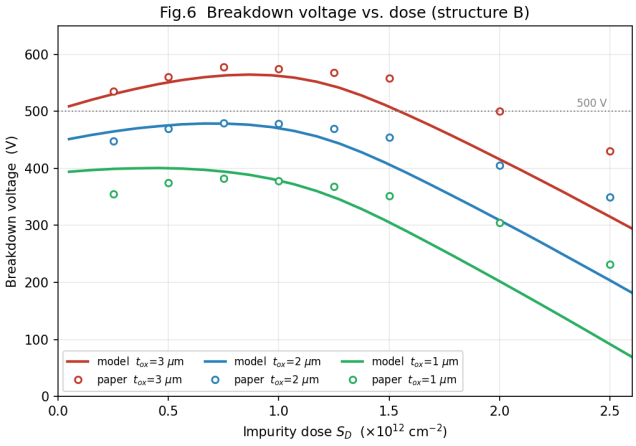


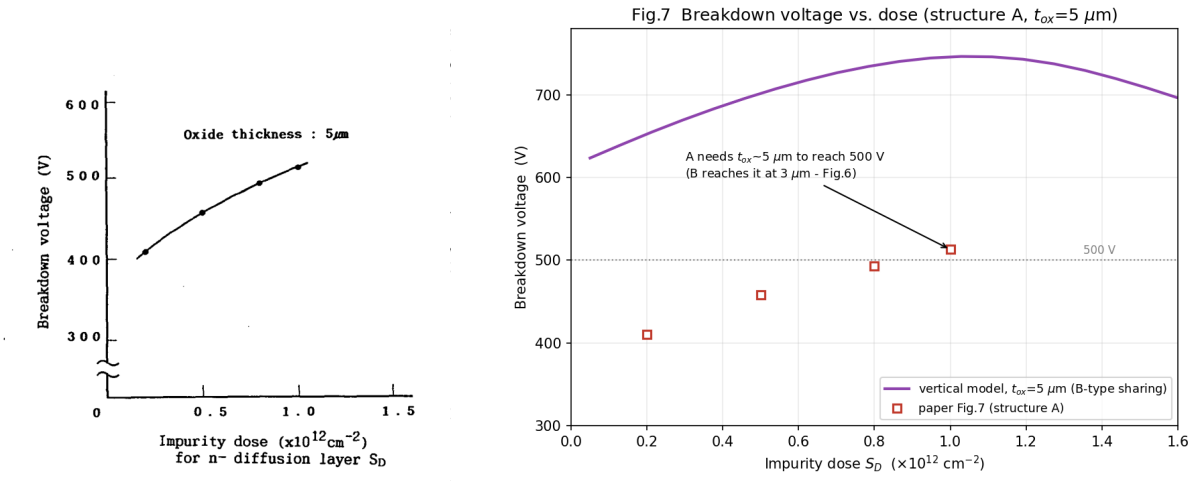
Fig.6 Breakdown voltage vs. impurity dose for shallow diffusion layer characteristics in the structure of Fig.3B.

(a) Paper (original)



(b) This reproduction

Figure 3: **Fig. 6 — breakdown voltage vs. dose (structure B).** Model curves (lines) vs. digitized paper points (circles) for $t_{\mathrm{ox}} = 1, 2, 3$ μm. The RESURF optimum — BV rising with dose, peaking near $1\times10^{12}\,\mathrm{cm}^{-2}$, then falling — is reproduced, with peak $BV \sim 400/480/565$ V vs. the paper's $\sim 380/480/580$ V.



(a) Paper (original)

(b) This reproduction

Figure 4: **Fig. 7 — breakdown voltage vs. dose (structure A, t_{ox} = 5 µm).** The vertical model (purple, B-type sharing) lies above the structure-A data because A and B share the same vertical stack — their difference is 2-D. The gap (∼150–200 V) is the signature of A's less complete field-sharing, consistent with A needing 5 µm oxide where B needs 3 µm.

3 Notes and limitations

What matches: the oxide carries >half the voltage (Fig. 4, model 54–60 %); the field shielding by the donor sheet (Fig. 5) with the correct ordering and $\sim 3\times$ interface jump; the $BV(S_D)$ RESURF curves and magnitudes (Fig. 6) within ~ 5 %.

Known limitation: structure A (Fig. 7) is a 2-D effect; the vertical model over-predicts its absolute BV. Reproducing it quantitatively requires the full 2-D device with the actual asymmetric drain/sinker doping.

Code: `nakagawa1990_full.py` (single-file PDE → plots), `model.py` (1-D electrostatics + avalanche), `solver2d.py` (2-D Poisson), `fig4--fig7_*.py` (figure scripts), `derivation.md` (full analytic derivation).